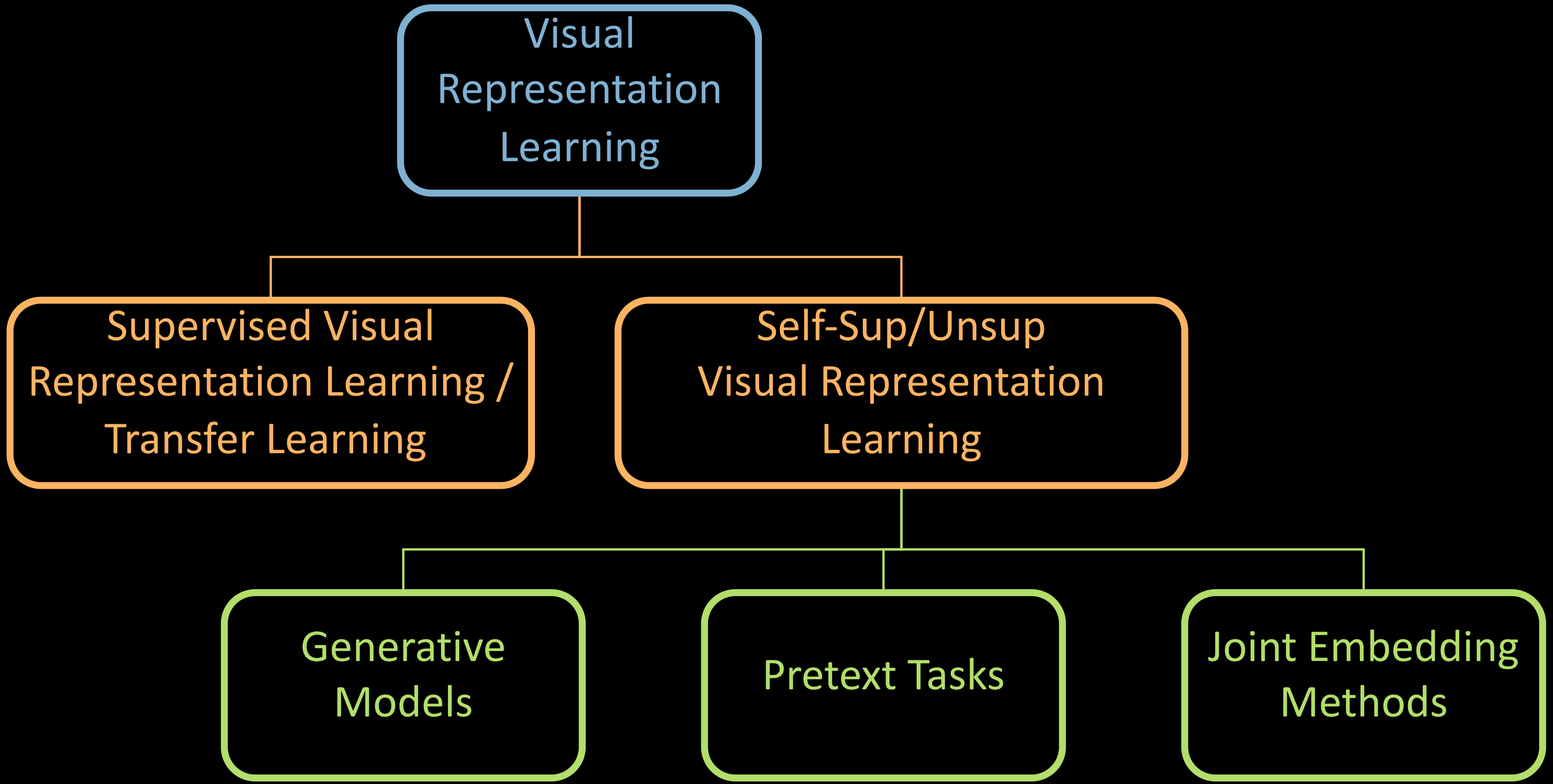
Joint Embedding Methods

Self-Supervised Visual Representation Learning

Visual Representation Learning



Self-supervised visual representation learning

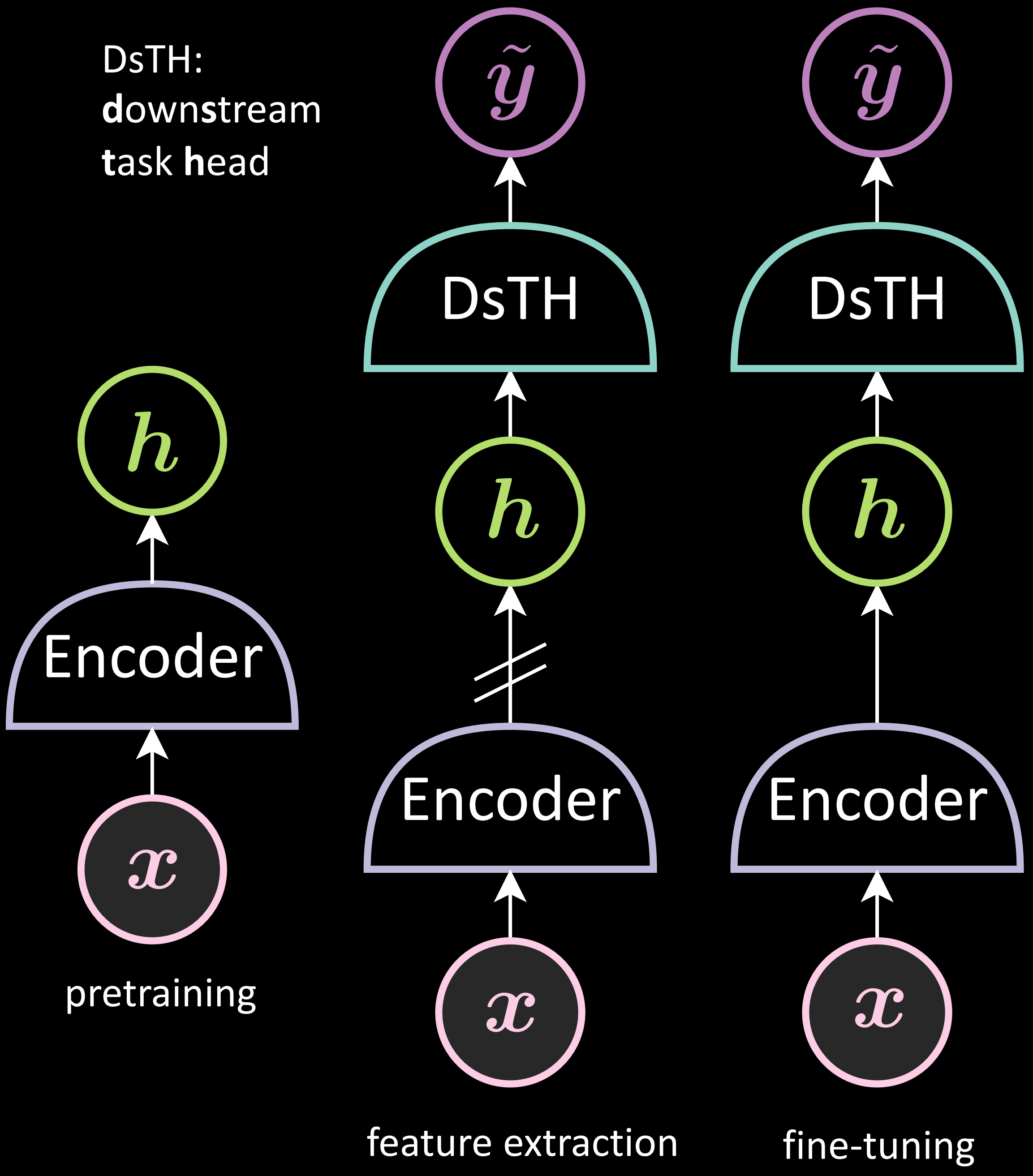
Step 1: pretraining

Use a large amount of unlabeled data to train a backbone network

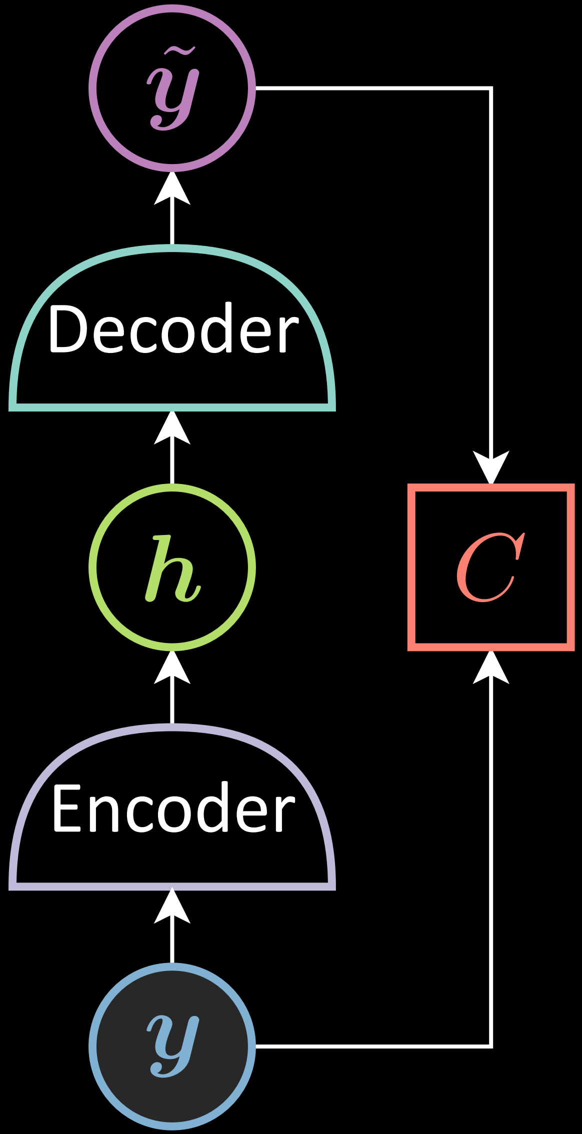
different methods will produce the backbone network differently

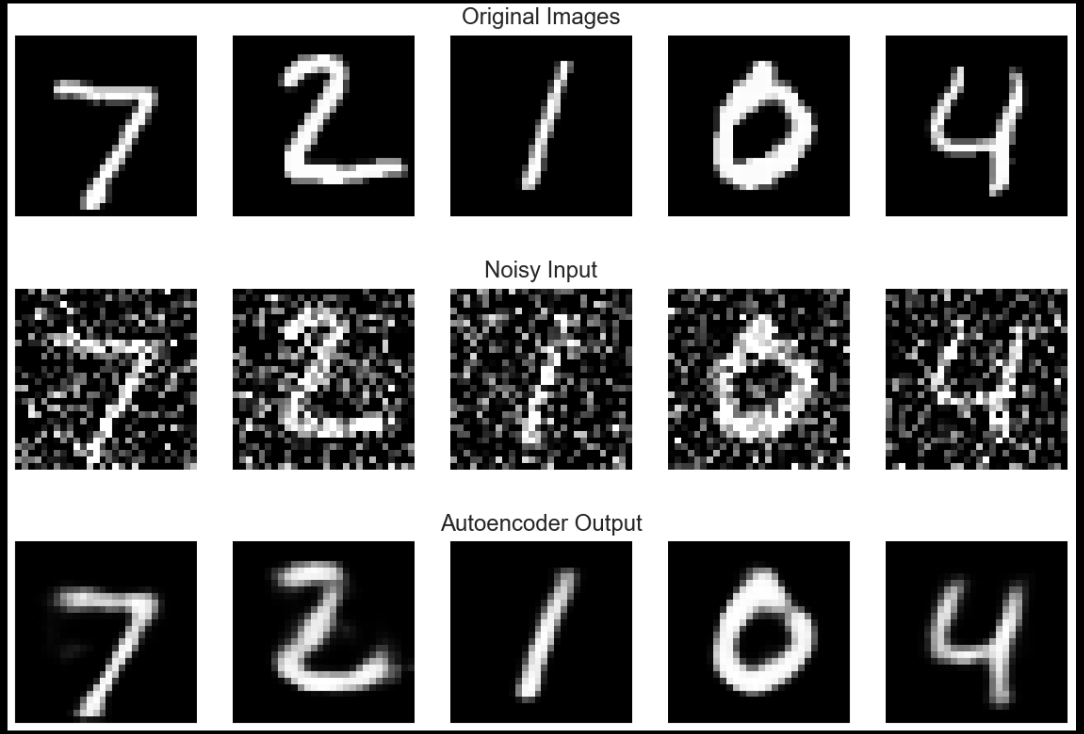
Step 2: evaluation

Use a small amount of labeled data to train a downstream task head network

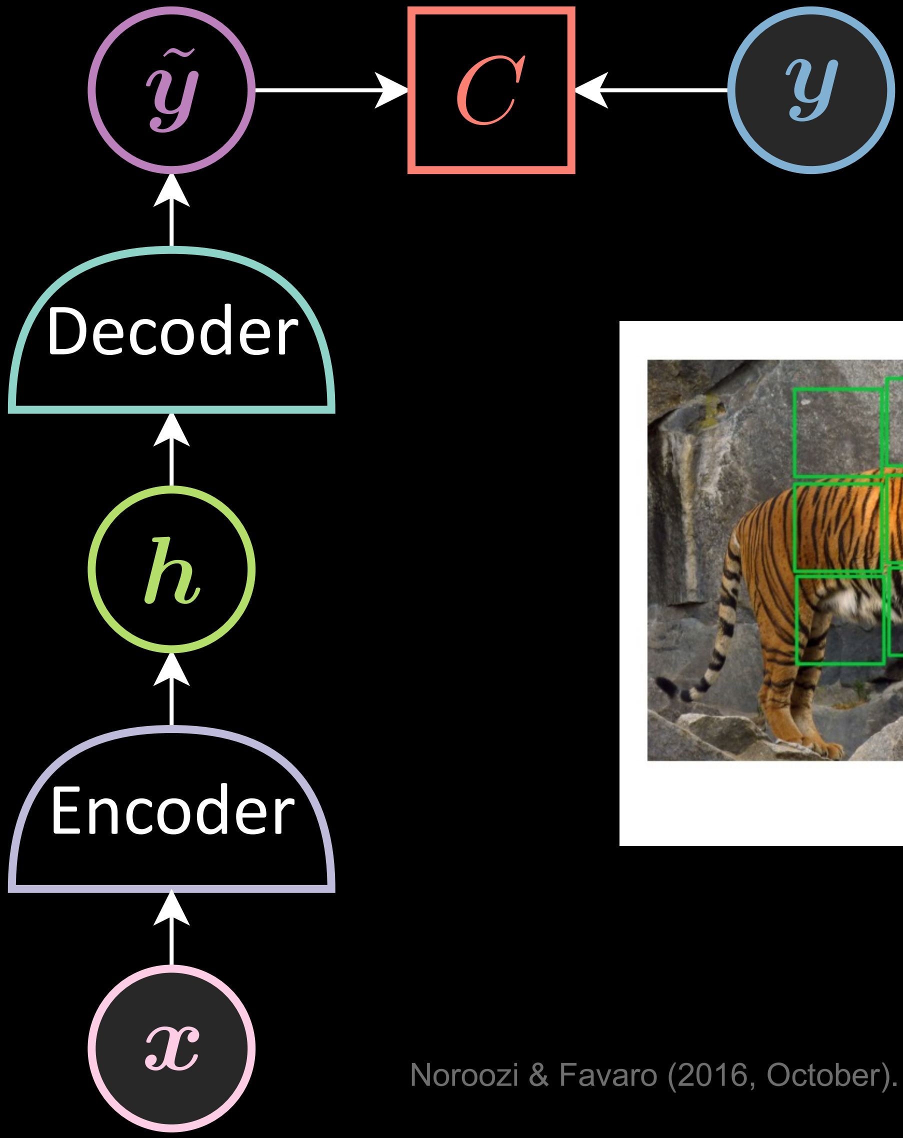


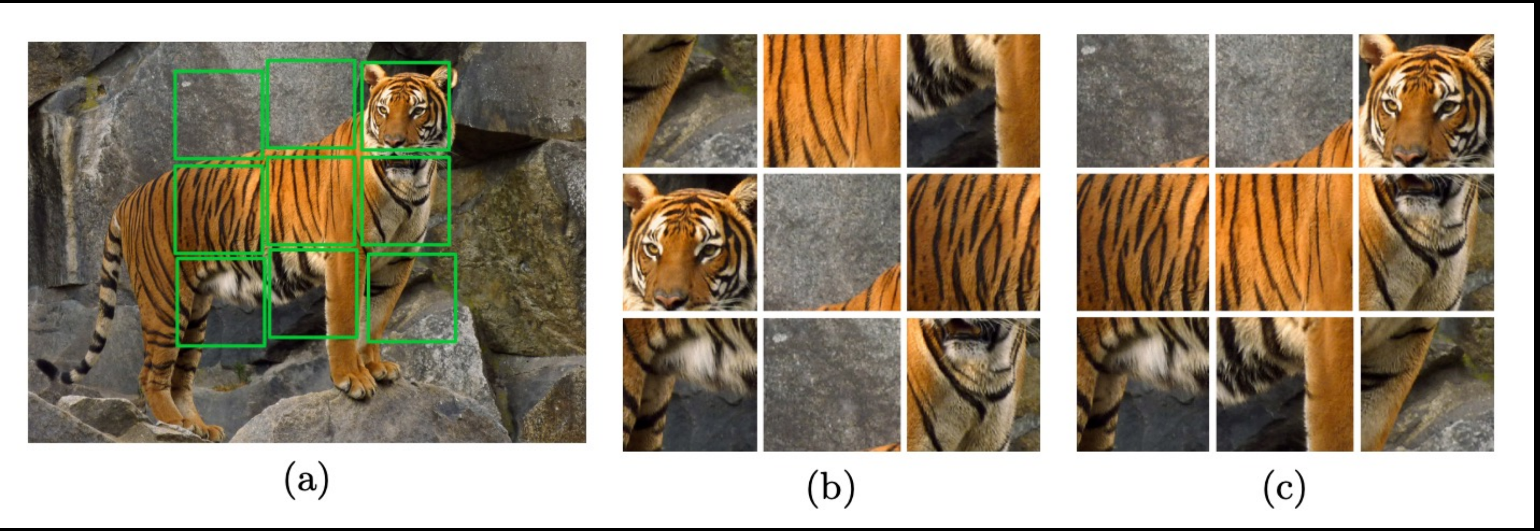
Generative Models - Autoencoder





Pretext Tasks

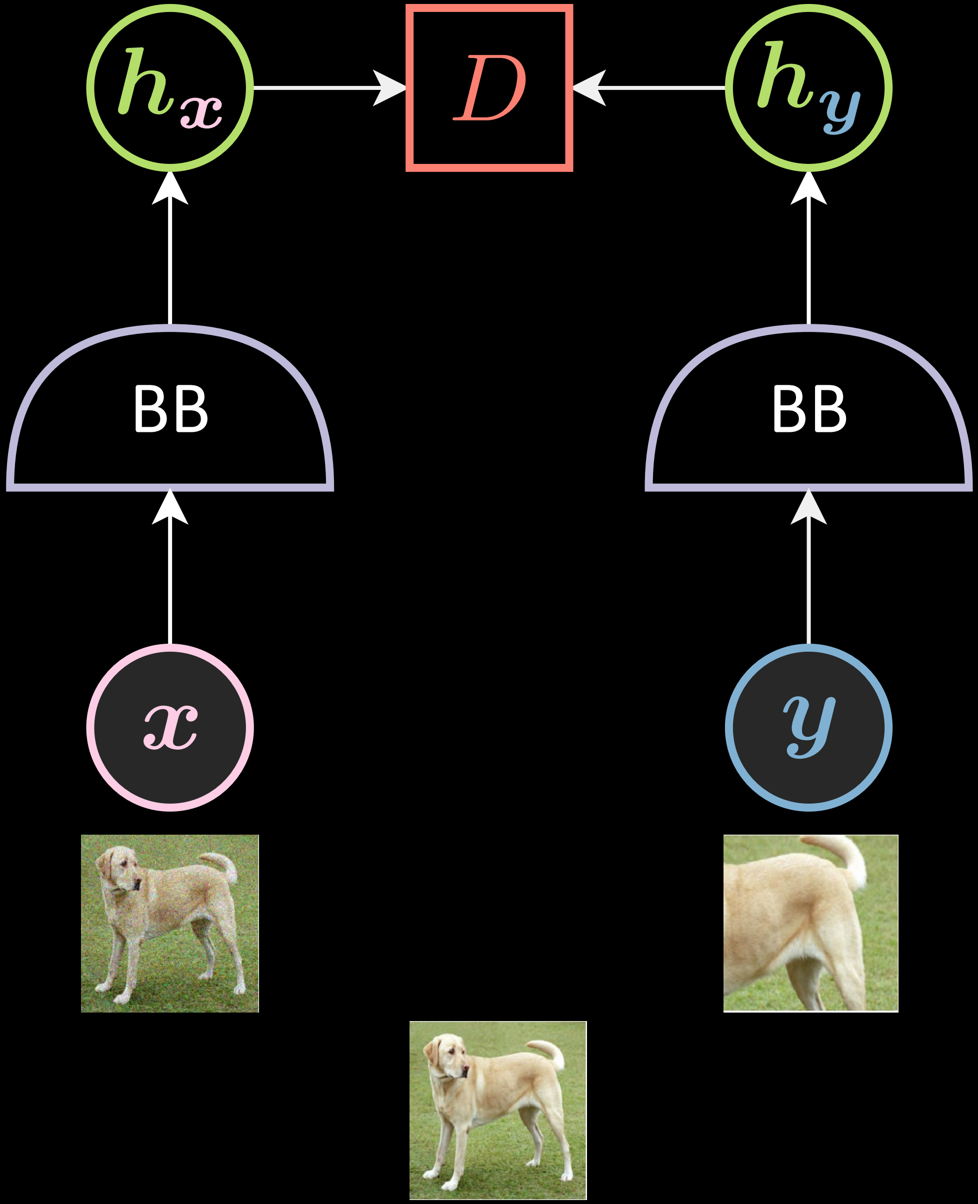




Noroozi & Favaro (2016, October). Unsupervised learning of visual representations by solving jigsaw puzzles.

Joint Embedding Methods

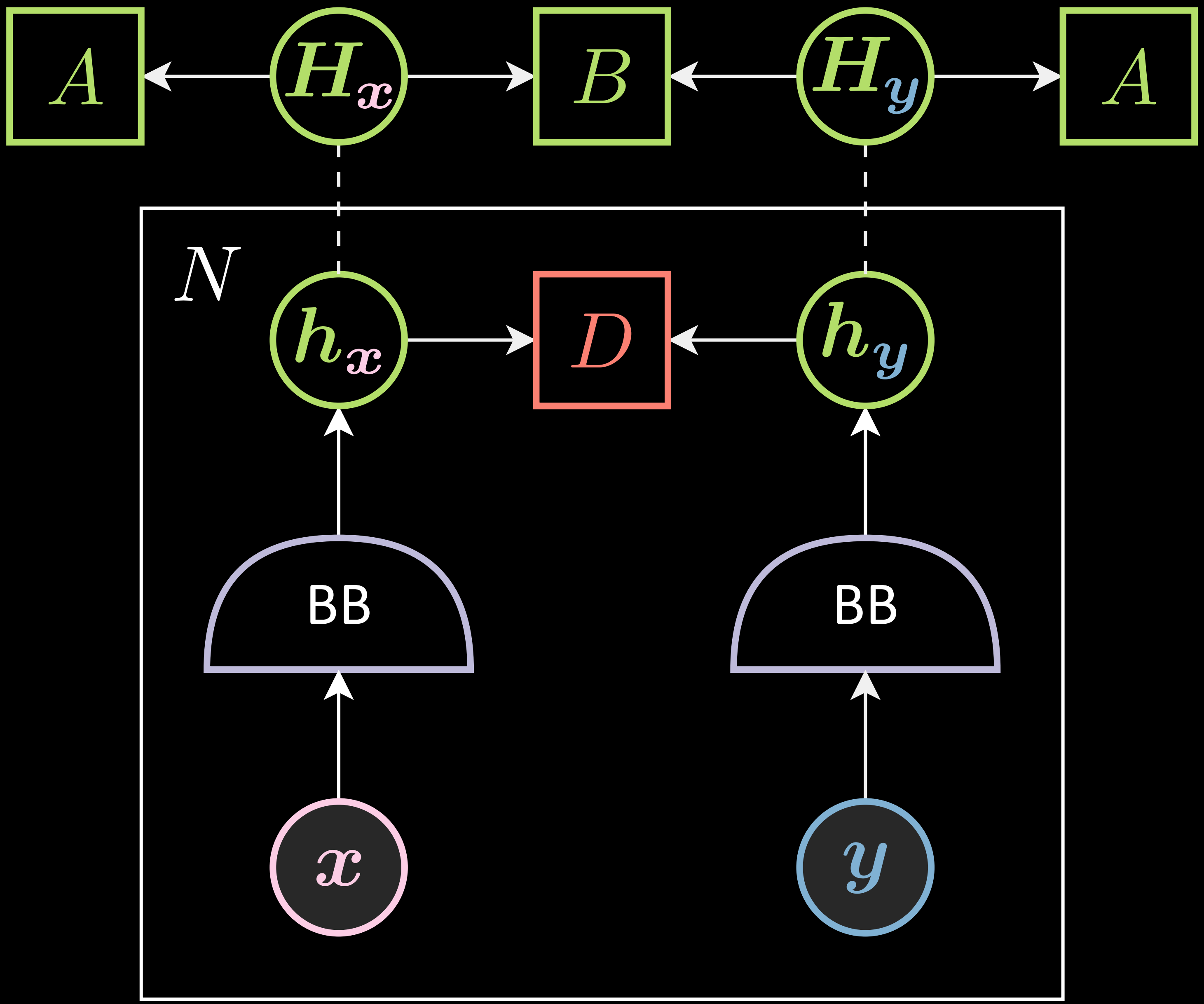
Good backbone network should be robust to certain distortions (invariant to data augmentation)



BB: backbone

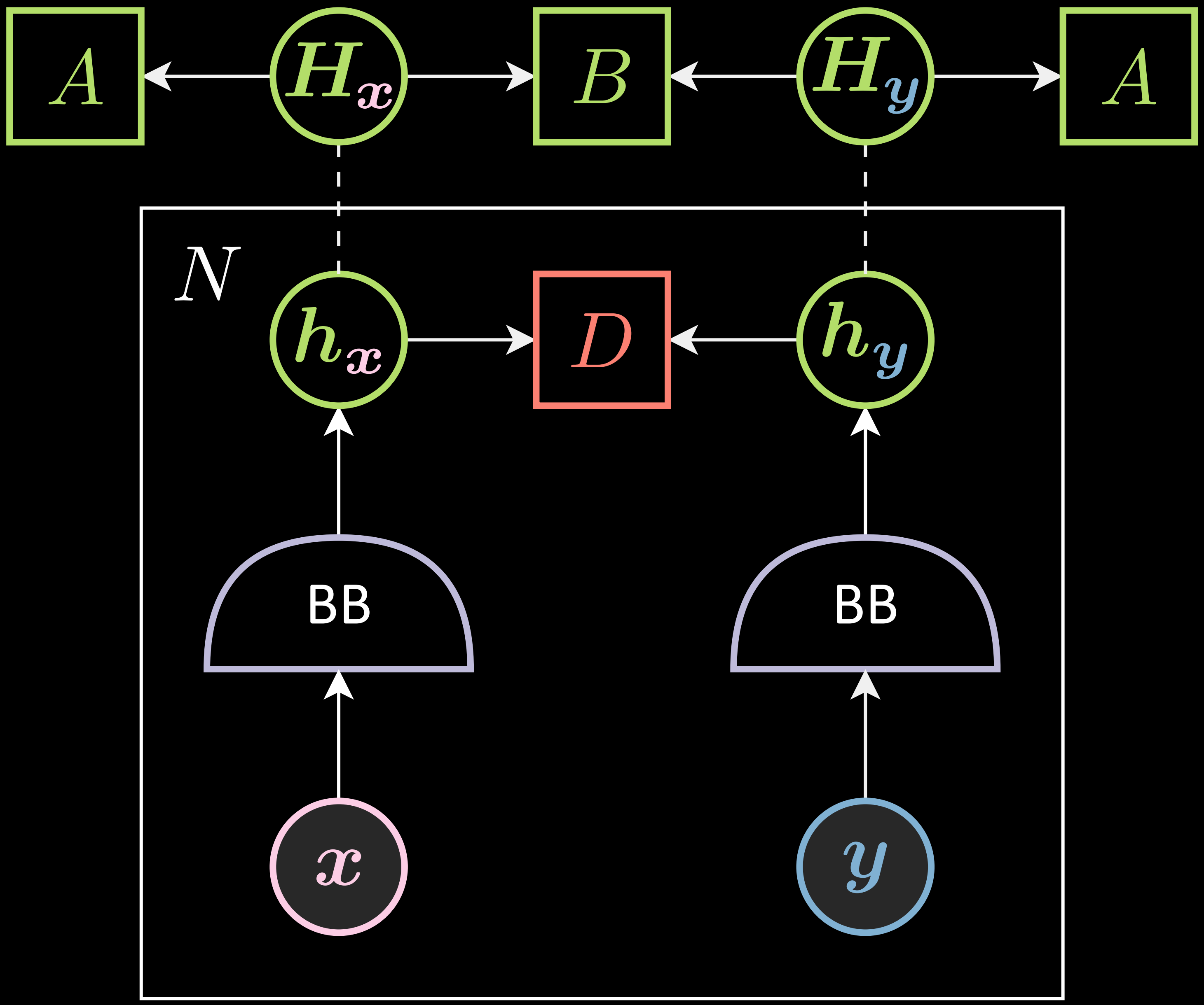
JEM

We add extra loss to prevent the trivial solution (constant embeddings)



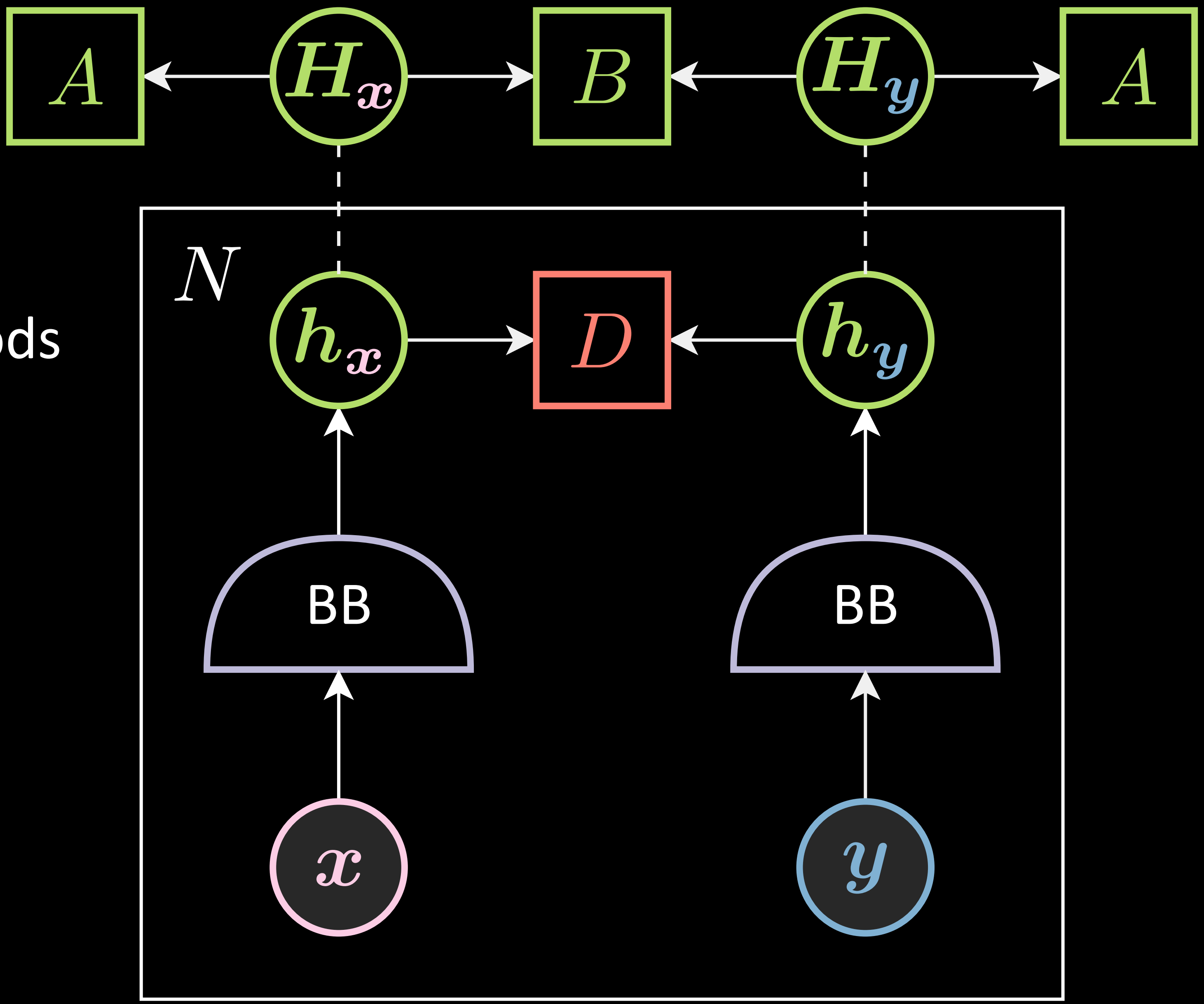
JEM

1. Data augmentation
2. Backbone network
3. Energy function
4. Loss functional



JEM

1. Contrastive methods
2. Non-contrastive methods
3. Clustering methods
4. Other methods



JEM loss functions

- A term that pushes the positive pair closer
- An (implicit) term that prevents the trivial solution (constant output)

To make the training stable, people usually normalize the embeddings or put a hinge on the loss function to prevent the norm of embeddings becoming too large or too small

Contrastive Methods

The loss function should push

1. the positive pairs closer

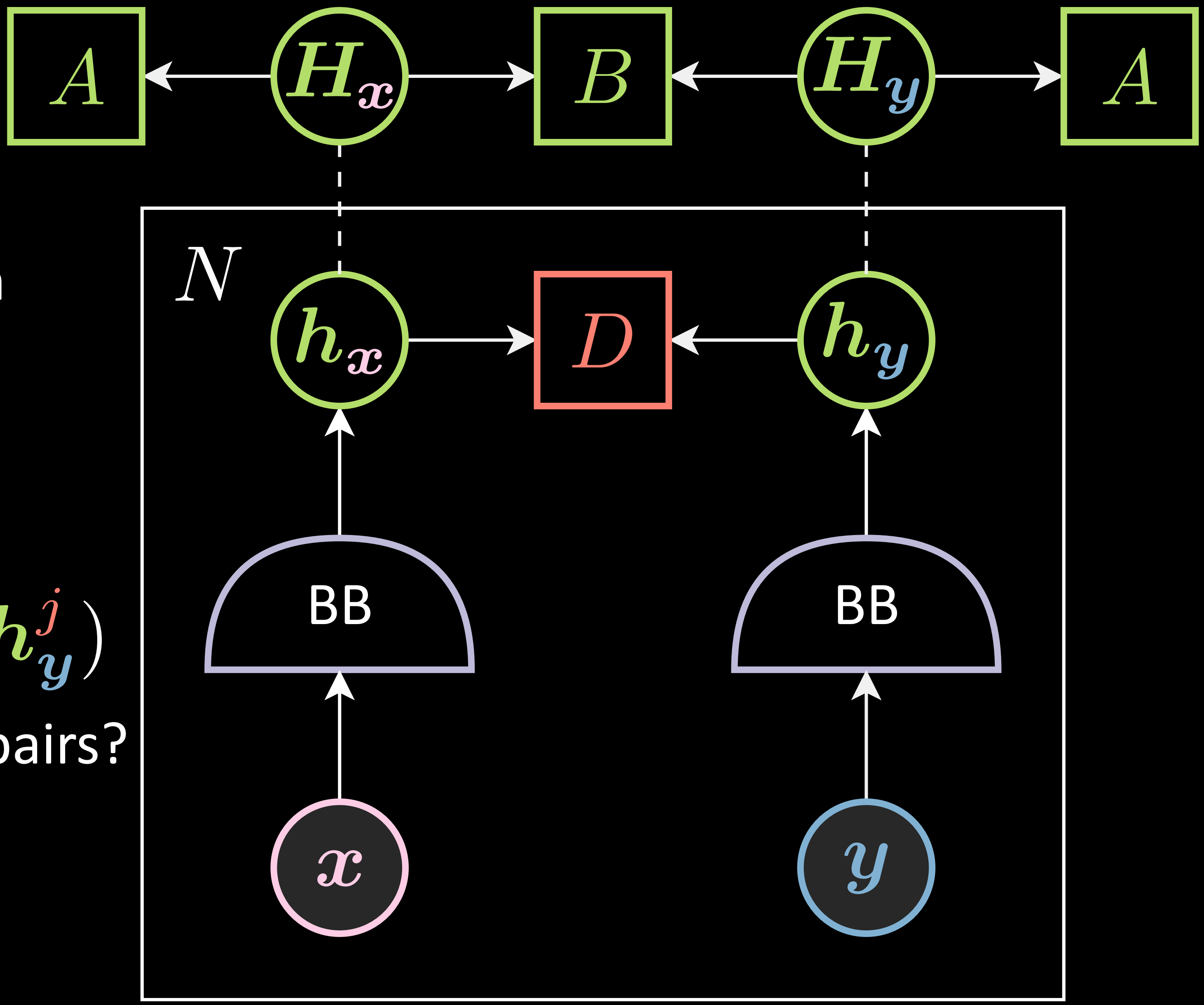
$$(\boldsymbol{h}_{\boldsymbol{x}}^{i}, \boldsymbol{h}_{\boldsymbol{y}}^{i})$$

2. the negative pairs away

$$(\boldsymbol{h}_{\boldsymbol{x}}^{i}, \boldsymbol{h}_{\boldsymbol{x}}^{j}), (\boldsymbol{h}_{\boldsymbol{x}}^{i}, \boldsymbol{h}_{\boldsymbol{y}}^{j}), (\boldsymbol{h}_{\boldsymbol{y}}^{i}, \boldsymbol{h}_{\boldsymbol{y}}^{j})$$

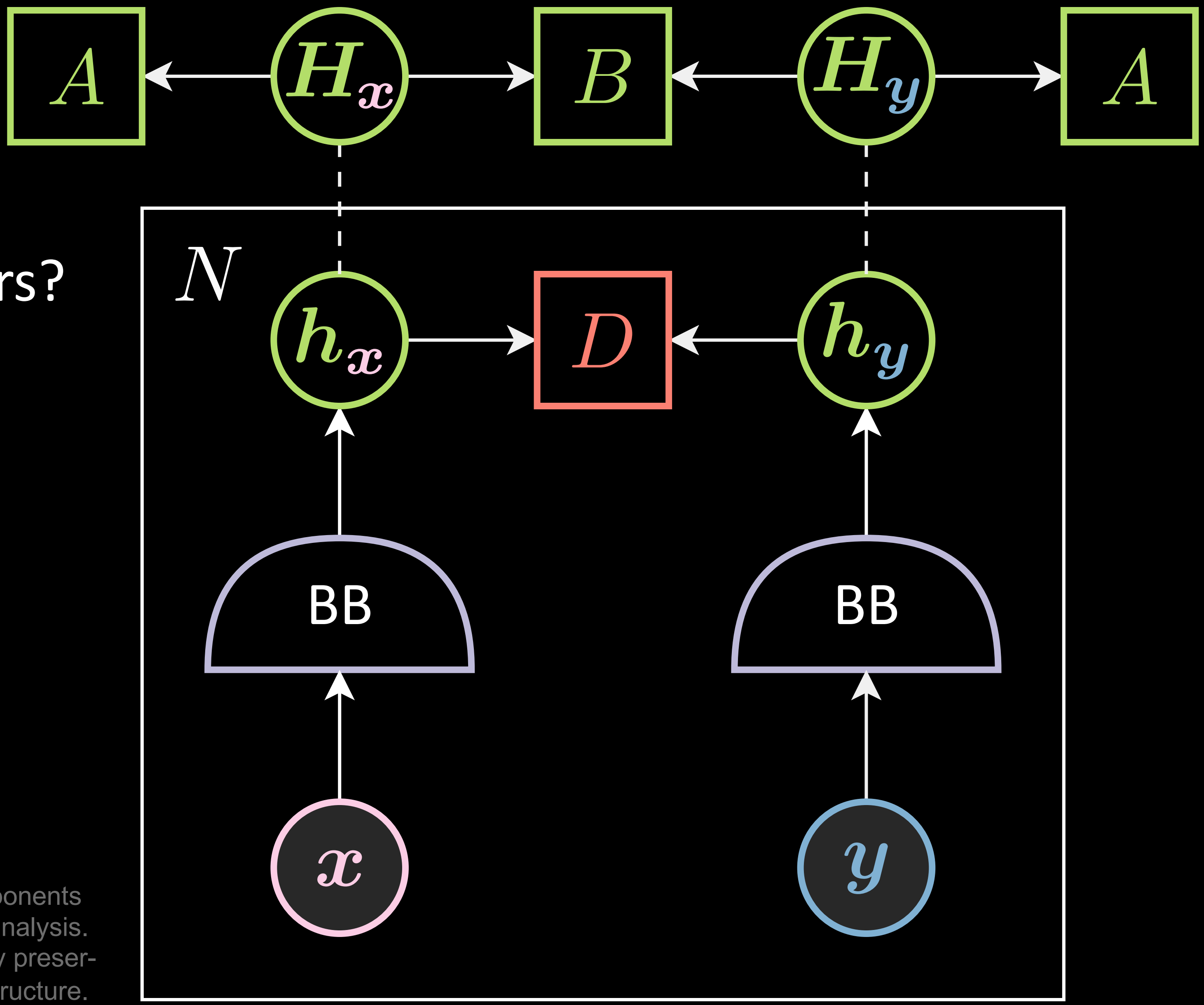
How to find a good negative pairs?

hard negative mining?



Chopra, Hadsell & LeCun (2005, June). Learning a similarity metric discriminatively, with application to face verification.
Hadsell, Chopra & LeCun (2006, June). Dimensionality reduction by learning an invariant mapping.

Contrastive Methods
SimCLR and MoCo



- How to find a good negative pairs?
- Use large batch size!

- Both SimCLR and MoCo use the InfoNCE loss function:

Goldberger, Hinton, Roweis & Salakhutdinov (2004). Neighbourhood components analysis.
Salakhutdinov & Hinton (2007, March). Learning a nonlinear embedding by preserving class neighbourhood structure.
Van den Oord, Li & Vinyals (2018). Representation learning with contrastive predictive coding.
Chen, Kornblith, Norouzi & Hinton (2020, November). A simple framework for contrastive learning of visual representations.
He, Fan, Wu, Xie & Girshick (2020). Momentum contrast for unsupervised visual representation learning.

The InfoNCE Loss Function

$$L(\boldsymbol{w}, \boldsymbol{x}, \boldsymbol{y}) =$$

$$= -\log \frac{\exp(\beta \, \mathrm{sim}(\boldsymbol{h_x}, \boldsymbol{h_y}))}{\sum_j^N \exp\left(\beta \, \mathrm{sim}(\boldsymbol{h_x}, \boldsymbol{h_x^j})\right) + \sum_j^N \exp\left(\beta \, \mathrm{sim}(\boldsymbol{h_x}, \boldsymbol{h_y^j})\right)}$$

$$= -\beta \, \mathrm{sim}(\boldsymbol{h_x}, \boldsymbol{h_y}) + \log \left[\sum_j^N \exp\left(\beta \, \mathrm{sim}(\boldsymbol{h_x}, \boldsymbol{h_x^j})\right) + \sum_j^N \exp\left(\beta \, \mathrm{sim}(\boldsymbol{h_x}, \boldsymbol{h_y^j})\right) \right]$$

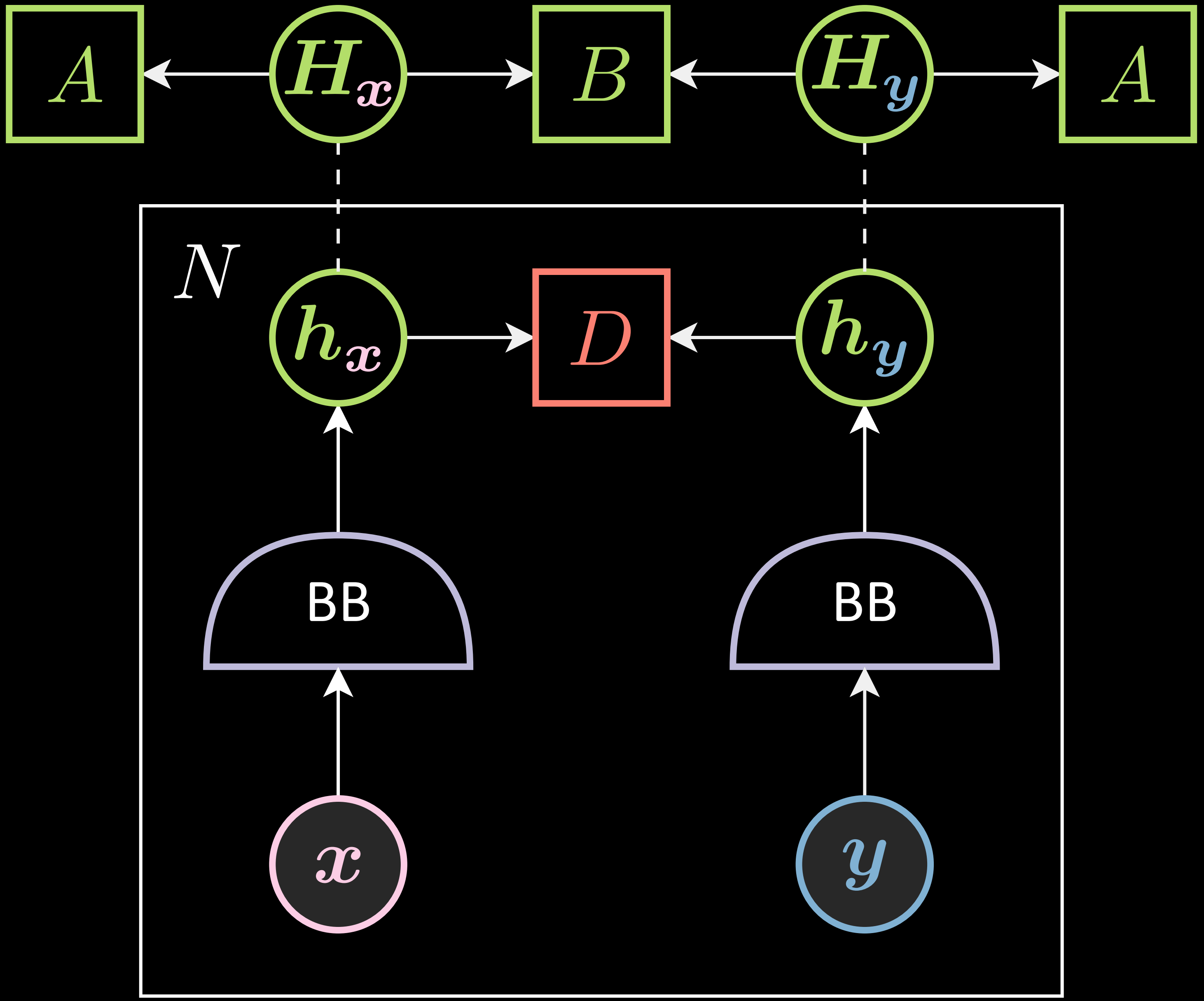
$$= -\beta \, \mathrm{sim}(\boldsymbol{h_x}, \boldsymbol{h_y}) + \underset{j}{\mathrm{softmax}_\beta} \left[\mathrm{sim}(\boldsymbol{h_x}, \boldsymbol{h_x^j}), \mathrm{sim}(\boldsymbol{h_x}, \boldsymbol{h_y^j}) \right]$$

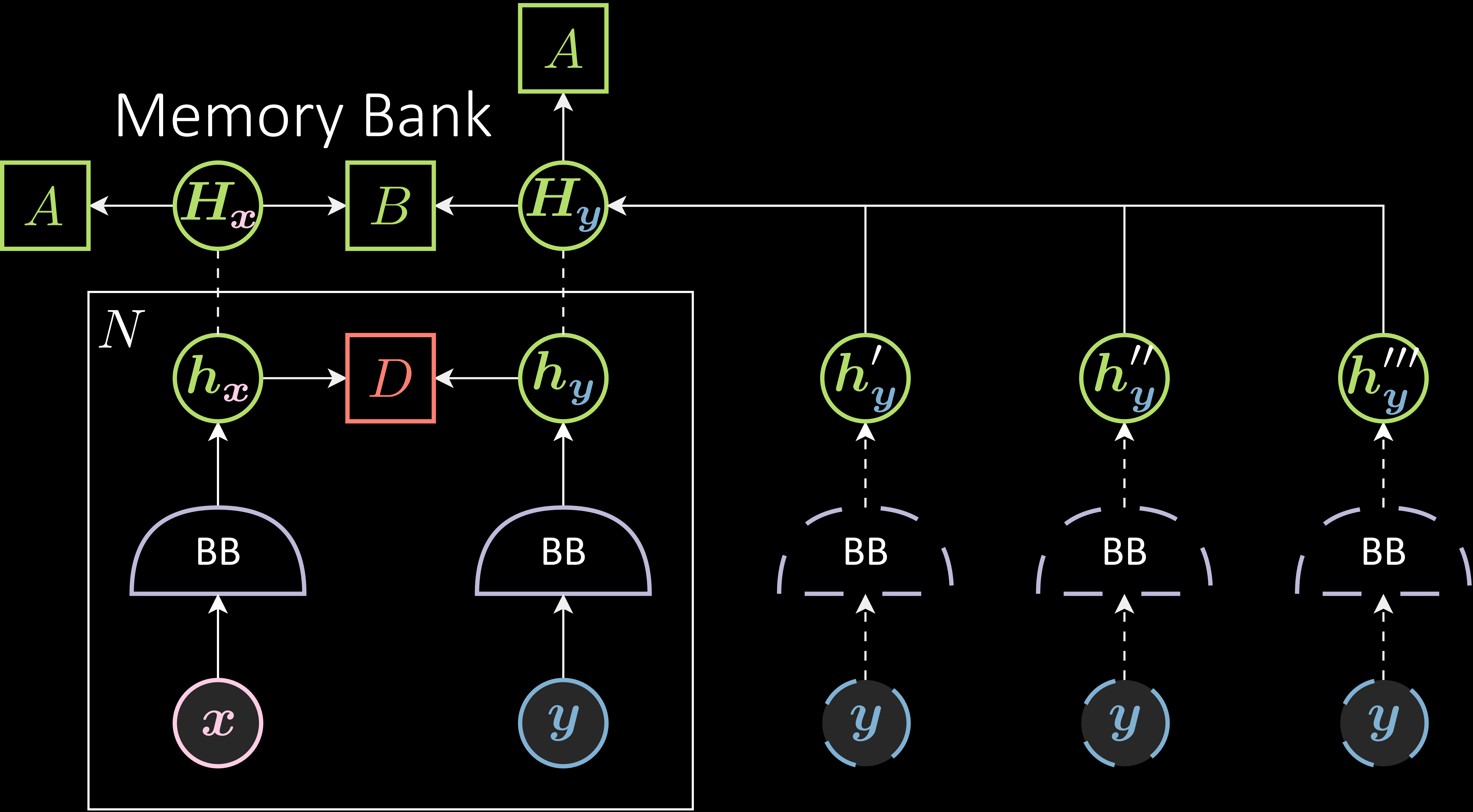
$$\mathrm{sim}(\boldsymbol{h_x}, \boldsymbol{h_y}) = \frac{\boldsymbol{h_x}^\top \boldsymbol{h_y}}{\|\boldsymbol{h_x}\| \|\boldsymbol{h_y}\|}$$

Batch size

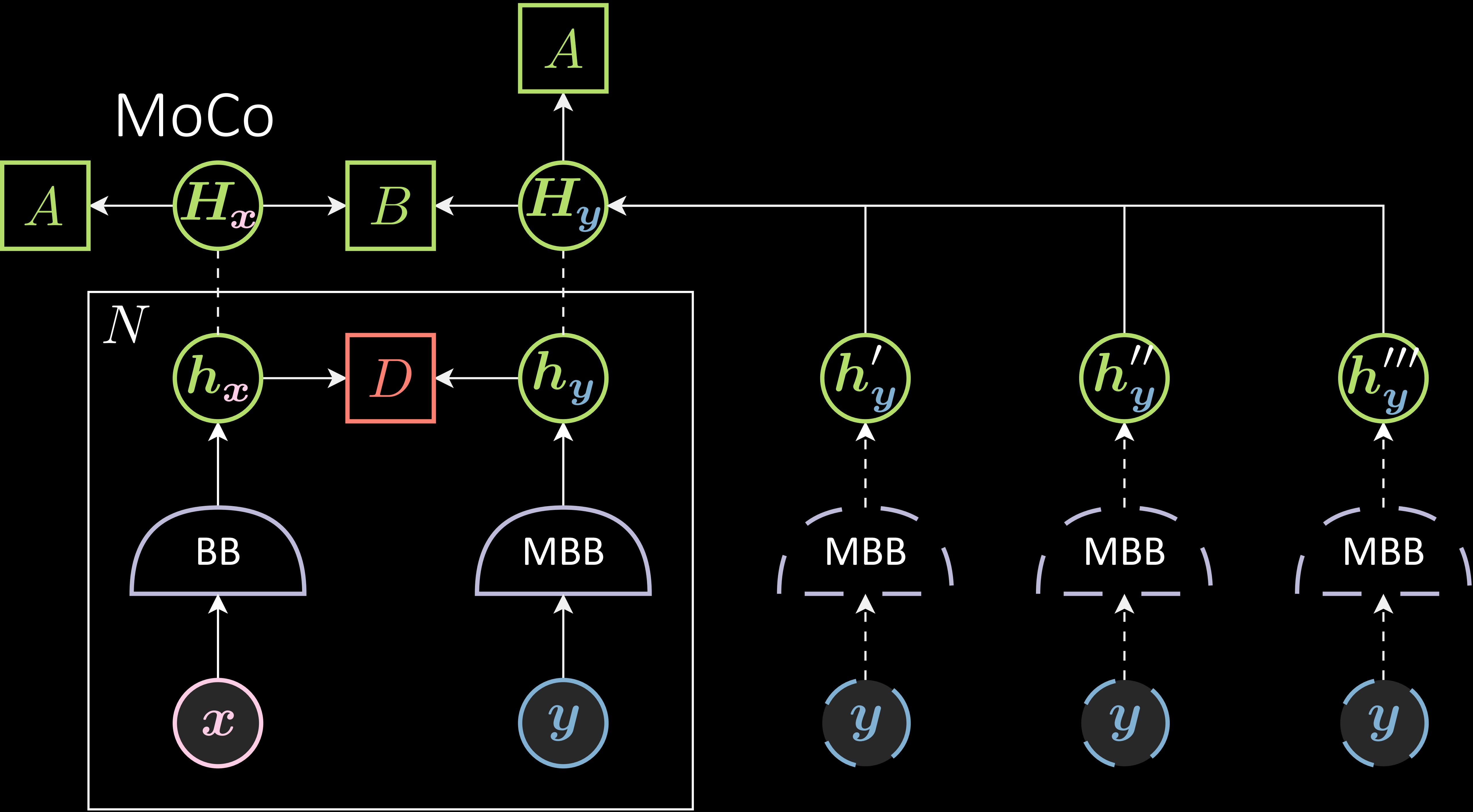
Very large!

SimCLR N = 8192



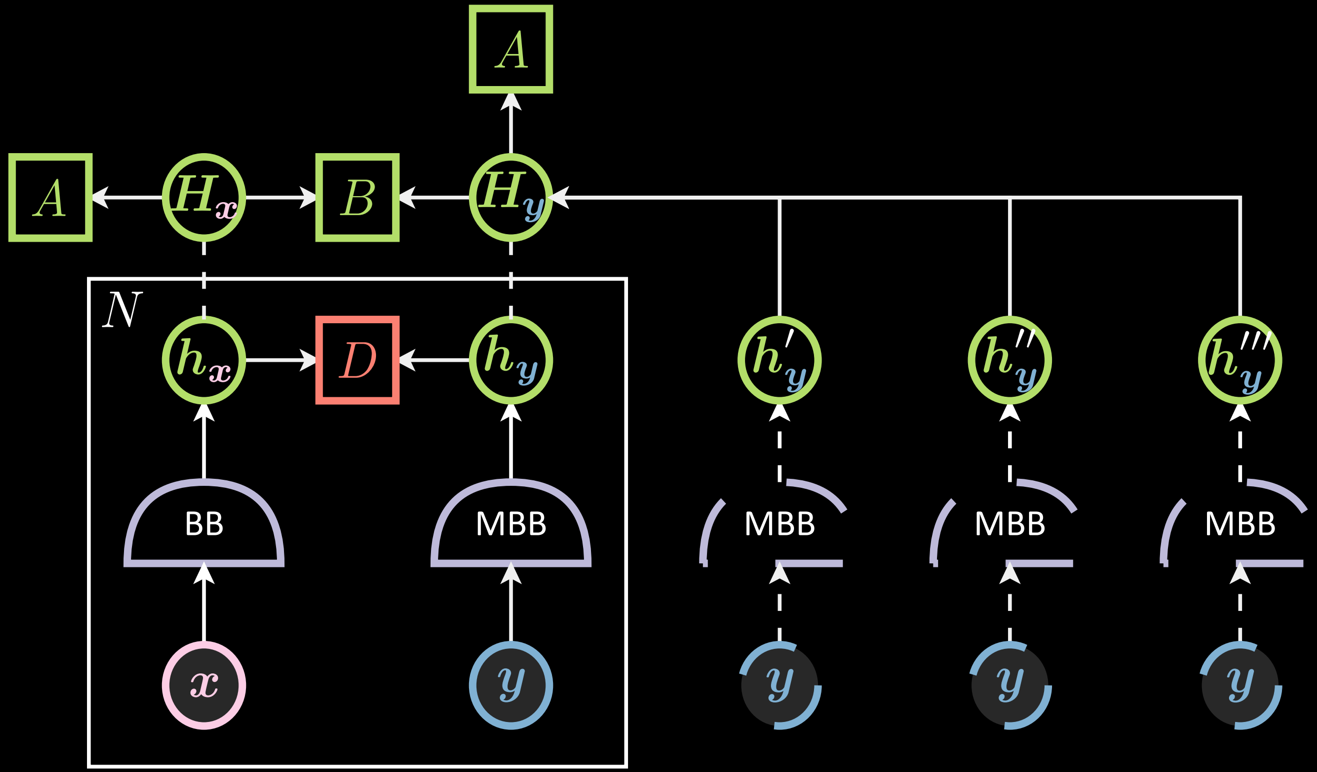


Memory Bank
A
A
H_x
B
H_y
N
h_x
D
h_y
h'_y
h''_y
h'''_y
BB
BB
BB
BB
BB
x
y
y
y
y



MoCo
A
H_x
B
H_y
A
N
h_x
D
h_y
h'_y
h''_y
h'''_y
BB
MBB
MBB
MBB
MBB
x
y
y
y
y

Contrastive Methods – MoCo



MoCo n = 256

$$\theta_{t+1} = \theta_t - \eta \nabla \theta_t$$

$$\vartheta_{t+1} = m\vartheta_t + (1 - m)\theta_{t+1}$$

θ :backbone parameters
ϑ :momentum backbone parameters